OTT SUBSCRIPTION CHURN ANALYSIS

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The Business Challenge

In the hyper-competitive OTT landscape (Netflix, Hotstar, Prime), retention is the only way to survive. You will act as a Data Analyst tasked with identifying high-risk subscribers using a multi-dimensional dataset (Customer demographics, Subscription tiers, and Support escalations).

Core Tech Stack

SQL & Python Integration (numpy, pandas, sqlite3, matplotlib, seaborn), Performance Engineering (data cleaning, feature engineering, analytics), Behavioral Visualization, Writing Actionable Insights

Project Milestones

• Relational Data Extraction: Connecting Python to SQL databases to pull multi-table datasets. • Advanced Feature Engineering: Data imports, calculating tenure, churn rates, and customer aging.

• Executive Reporting: Translating technical findings into billion-dollar business insights.

Note: This isn't just a coding exercise, it’s a business solution. You’ll leave with a portfolio-ready project applicable to E-commerce, SaaS, Fintech and Adtech industries.

ROADMAP

1. Connect SQL database to Python – pandas and sqlite3
2. Data import using SQL query in Python – pandas and sqlite3
3. Data Cleaning – numpy and pandas
    - data types, rename cols, select specific cols, QCs, handle missing/null values
4. Feature Engineering – numpy and pandas
    - Create new calculated cols, data transformation, use filters
5. Data Analysis – numpy and pandas
    - EDA - aggregation, group by, pivot table
6. Data Visualization – matplotlib and seaborn
7. Insights on ppt – key insights, next steps, impact to show in resume

SUMMARY

Engineered an end-to-end churn analytics pipeline for an OTT subscription dataset (20+ KPIs), segmented the customer base into risk tiers using composite churn scores, exposed a significant **lifetime** value gap between churned and retained cohorts, and recommended prioritising Premium annual-plan retention over Basic monthly acquisition to maximise long-term revenue yield. Identified a 28.6% overall churn rate. The retention rate is around 71.4%.

CHURN DEFINITION BY BUISNESS

Churn analysis is the process of figuring out why customers stop doing business with you. For this, we only need to look at three things:

- "Who": Identifying which customers left or will leave in future
- "Why": Analyzing their behavior before they left.
- "When": Finding the "danger zone" – point when a user leave

Business Type	**Churn Definition**
SaaS	Subscription canceled
E-commerce	No purchase in 90 days
Adtech	Using service/app in 90 – 120 days
Streaming	Membership inactive
Telecom	Account terminated
Banking	No transactions for X months

DATABASE TABLES

db_customer
customerid
name
country
State
gender
dob
interests
pincode

db_support
customerid
complaint_date
Escalations
csat_score
col_1
comment

db_subscription
customerid
subscription_start_date
subscription_type
renewal_date
plan_type
contract_type
cancellation_date
cancellation_reason
monthly_charges
cltv
churn_score

CALCULATED METRICS

KPI	Formula
churn rate	churned customer / total customers
Churn by plan type	churn rate GROUP BY plan_type (Basic/Standard/Premium)
Churn by state	churn rate GROUP BY country, state
Retention rate	1-churn rate
ARPU	SUM(monthly_charges) / COUNT(active customerid)
Average customer tenure	AVG(DATEDIFF(cancellation_date OR NOW(), subscription_start_date))
Revenue at risk	SUM(monthly_charges) WHERE churn_score > 70
Escalation rate	SUM(escalations) / COUNT(complaints) × 100
Avg complaints per customer	COUNT(complaints) / COUNT(DISTINCT customerid)
Correlation: escalations → churn	churn rate WHERE escalations ≥ 1 vs 0

INSIGHTS

Insights

- Churn Rate: 28.6% | Retention Rate: 71.4%
- Most of the churn is from basic subscription plan – nothing to worry in terms of major revenue impact
- Most of the churn happened in the month of Sep 2024 and, most affected state is Karnataka
- Average Tenure (Days): 1,451 | ARPU is Rs 18.8
- Total Revenue = 395
- Revenue loss due to churn = 74
- % Revenue loss = 18%
- monthly vs annual churn = 55.6% vs 8.3%

Action Items

- Check what happened in Karnataka – was there any increase in price, user complaints, tech issues, etc
- Did we increased subs price for **Basic plan** or recently made any changes – specially in sep month?
- Check what competitor is doing – bcoz one user has moved to competitor
- Focus on customers with 'High' & 'Med' churn risk, check their LTV (to make priority list), complaint request, reach out to them through email, sms, calls and fix their issue.